

AZURE STADIUM TICKET BOOKING INFRASTRUCTURE

Cloud Infrastructure Project using Microsoft Azure

Registration Number: 12309110

Section: K23MB

Submitted By: Sunil Ganganagunta

Azure for Students | Microsoft | CreateVm-microsoftwindowsserver2022-20260414223607 | Overview

portal.azure.com/#view/HubsExtension/DeploymentDetailsBlade/~/overview/id/%2Fsubscriptions%2F32ca749f-f8b6-42b6-...

Microsoft Azure

Home

CreateVm-microsoftwindowsserver.windowsserver2022-20260414223607 | Overview

Deployment

Search

Overview

Inputs

Outputs

Template

✓ Your deployment is complete

Deployment name : CreateVm-microsoftwindowsserver.windowsserver2022-2026041422... Start time : 14/04/2026, 22:37:25
Subscription : Azure for Students Correlation ID : 91817c59-b539-42a4-b2c5-820fc1c5580e
Resource group : rg-medicloud-dr-sg

Deployment details

Next steps

Set up auto-shutdown Recommended
Monitor VM health, performance, and network dependencies Recommended
Run a script inside the virtual machine Recommended

Go to resource Create another VM Scale out your VM

Cost management
Get notified to stay within your budget and prevent unexpected charges on your bill.
[Set up cost alerts >](#)

Battery status: 58% remaining

Search 29°C Partly cloudy ENG IN 22:39 14-04-2026

Azure for Students | Microsoft | Microsoft.RecoveryServicesV2-1776186824862 | Overview

portal.azure.com/#view/HubsExtension/DeploymentDetailsBlade/~/overview/id/%2Fsubscriptions%2F32ca749f-f8b6-42b6-...

Microsoft Azure

Home

Microsoft.RecoveryServicesV2-1776186824862 | Overview

Deployment

Search

Overview

Inputs

Outputs

Template

✓ Your deployment is complete

Deployment name : Microsoft.RecoveryServicesV2-1776186824862 Start time : 14/04/2026, 22:43:48
Subscription : Azure for Students Correlation ID : 5931c765-0682-4afa-b0ba-5fa42ba91d75
Resource group : rg-medicloud-dr-sg

Deployment details

Next steps

Go to resource

Cost management
Get notified to stay within your budget and prevent unexpected charges on your bill.
[Set up cost alerts >](#)

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14 April 2026
Tue 22:45 (Local time)

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Microsoft Azure

Home > Compute infrastructure

Compute infrastructure | Virtual machines

Search resources, services, and docs (G+/)

Copilot

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DEFAULT DIRECTORY (SUNILGAN...)

Home > Compute infrastructure

Virtual machines

Get started

Filter for any field...

Subscription equals all Type equals all Resource Group equals all Location equals all Add filter

Name	Subscription	Resource Group	Location	Status	Operating syst...	Size	Public IP addre...	Disks	Update status
VM1-AzureLab	Azure for Stude...	RG-AzureLab	Central India	Running	Windows	Standard_B2ats...	20.40.41.206	1	Enable periodic...
VNet-AzureLab	Azure for Stude...	RG-AzureLab	Central India	Running	Windows	Standard_B2ats...	4.186.30.122	1	Enable periodic...

Overview

All resources

Infrastructure

Virtual machines

Virtual Machine Scale Set (VMSS)

Compute Fleet

Arc Machines

Disks + images

Capacity + placement

Related services

Monitoring + Operations

Help

Microsoft Azure

Home > Network foundation

Network foundation | Virtual networks

Search resources, services, and docs (G+/)

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Home > Network foundation

Virtual networks

Suggest a firewall SKU for my network topology Summarize vNets by location Recommend connectivity for listed networks

Filter for any field...

Subscription equals all Resource Group equals all Location equals all Add filter

Name	Resource Group	Location	Subscription
VM1-AzureLab-vnet	RG-AzureLab	Central India	Azure for Students
VNet-AzureLab-vnet	RG-AzureLab	Central India	Azure for Students
vnet-lab	rg-vm-lab	Central India	Azure for Students

Overview

Virtual network

Virtual Network overview

Virtual networks

NAT gateways

Public IP addresses

Network interfaces

Network security groups

Application security groups

Bastions

Route tables

Route servers

Private Link

DNS

Monitoring and management

Search

33°C Sunny

ENG IN

10:52

07-05-2026

Microsoft Azure

Home > Network foundation | Virtual networks

Network foundation | Virtual networks

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Virtual networks

Retrieve detailed routing information for troubleshooting Evaluate routing for this virtual network +1

Filter for any field...

Subscription equals all Resource Group equals all Location equals all Add filter

VM1-AzureLab-vnet

Virtual network

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Monitoring

Automation

Help

Essentials

Resource group (move)

Address space

10.1.0.0/16

Subnets

1 subnet

DNS servers

Azure provided DNS service

BGP community string

Configure

Virtual network ID

287ea861-a55b-4a45-a921-bd6f0f07e038

Tags (edit)

Add tags

Topology Properties Capabilities Recommendations Tutorials

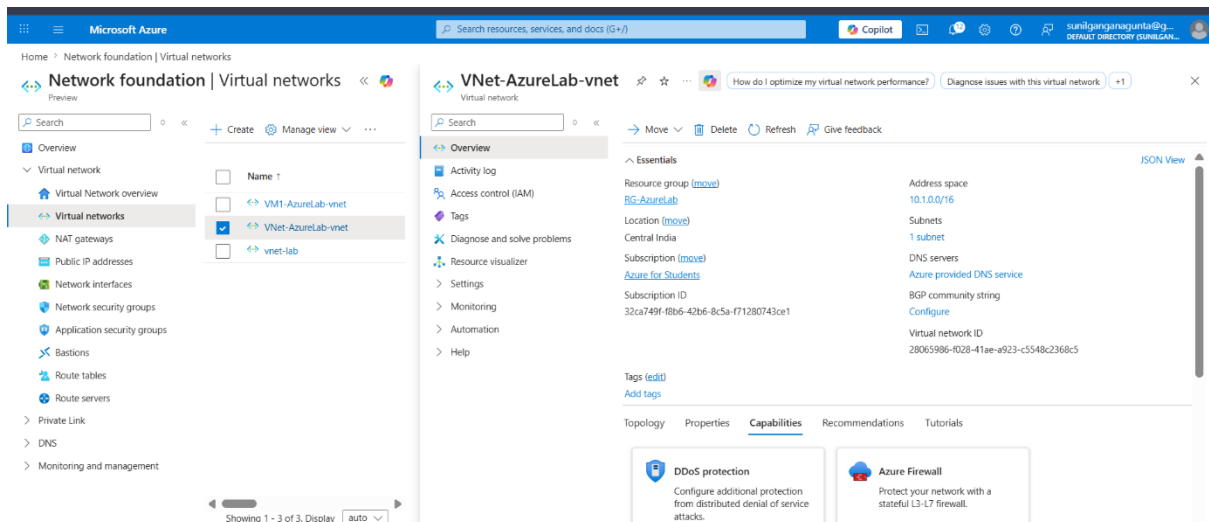
DDoS protection

Configure additional protection from distributed denial of service attacks.

Azure Firewall

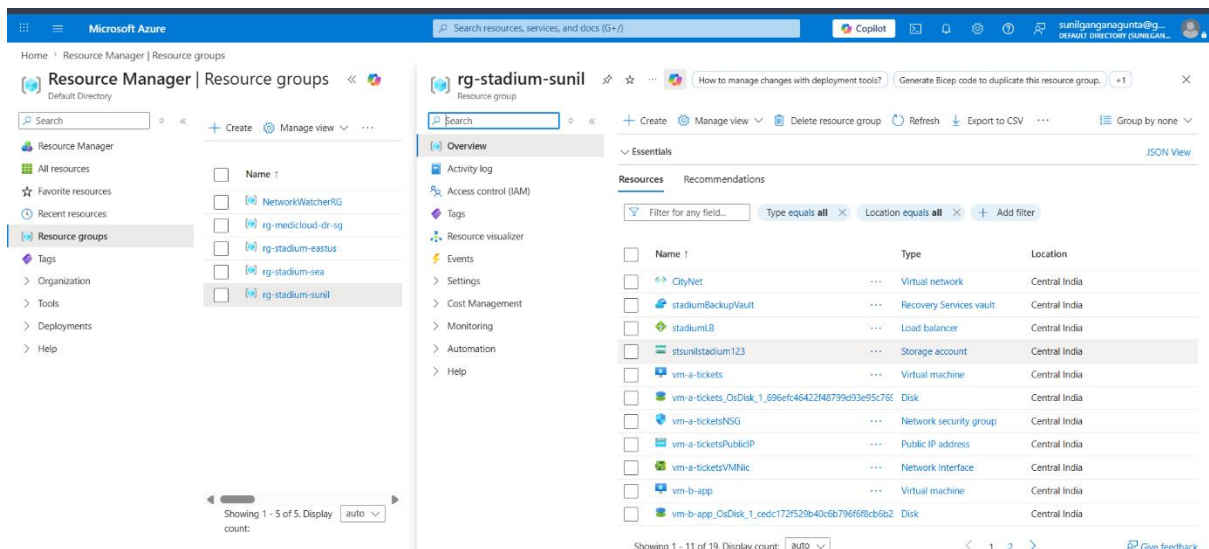
Protect your network with a stateful L3-L7 firewall.

Showing 1 - 3 of 3. Display count: auto



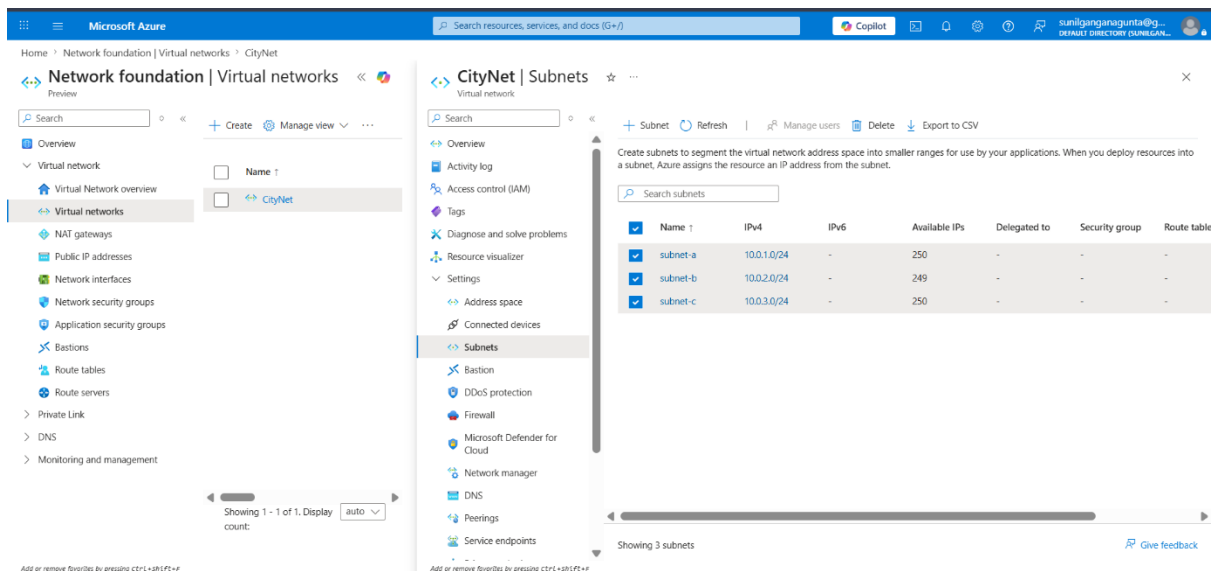
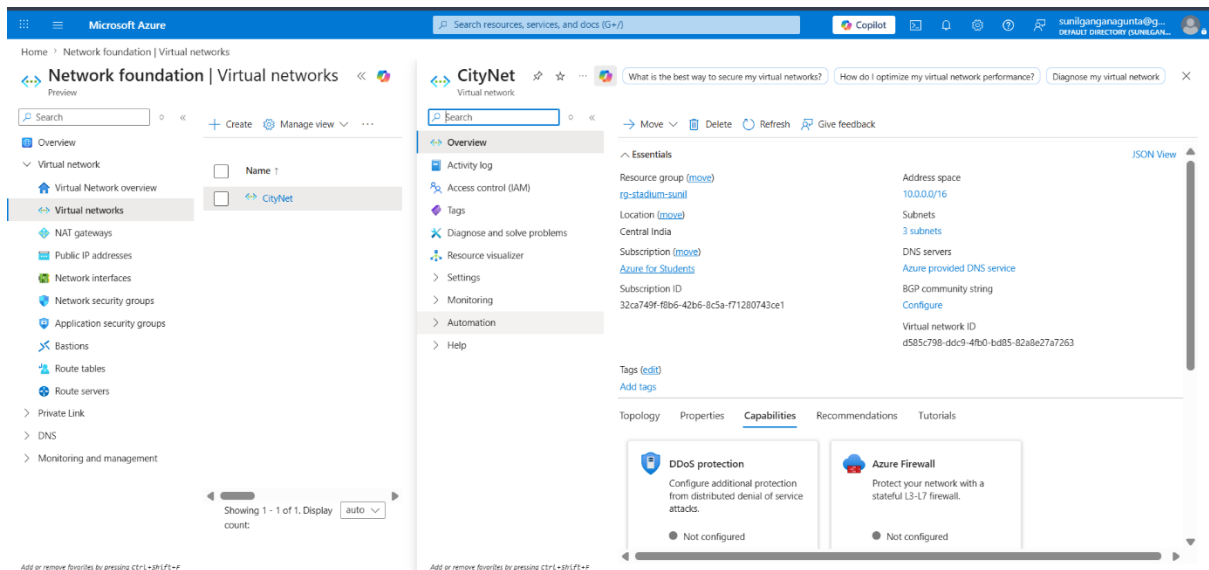
2. RESOURCE GROUP :-

Resource Groups → rg-stadium-sunil



3. VIRTUAL NETWORKS :-

Virtual Networks → CityNet



4. VIRTUAL MACHINES :-

Three Ubuntu Virtual Machines were deployed to simulate a multi-tier cloud application architecture. VM-A acts as the Ticketing Server, VM-B acts as the Application Server, VM-C acts as the Broadcast Server.

Microsoft Azure

Home > Compute infrastructure

Compute infrastructure | Virtual machines

Search resources, services, and docs (G+/I)

Identify VMs with network connectivity issues | Check which VMs cannot access the internet | Get workload templates for VMs

Search

Overview | All resources | Infrastructure | Virtual machines

Virtual Machine Scale Set (VMSS) | Compute Fleet | Arc Machines

Disks + images | Capacity + placement | Related services | Monitoring + Operations | Help

+ Create | Reservations | Manage view | Refresh | Export to CSV | Open query | Assign tags | Start | Restart | Stop | Delete | Services | Group by none

Filter for any field... | Subscription equals all | Type equals all | Resource Group equals all | Location equals all | Add filter

Name	Subscription	Resource Group	Location	Status	Operating syst...	Size	Public IP addre...	Disks	Update status
vm-a-tickets	Azure for Stude...	rg-stadium-sunil	Central India	Running	Linux	Standard_B2ats...	20.198.112.99	1	Enable periodic...
vm-b-app	Azure for Stude...	rg-stadium-sunil	Central India	Running	Linux	Standard_B2ats...	20.204.102.238	1	Enable periodic...
vm-c-broadcast	Azure for Stude...	rg-stadium-sunil	Central India	Running	Linux	Standard_B2ats...	20.219.219.217	1	Enable periodic...

Showing 1 - 3 of 3. Display count: auto

Add or remove favorites by pressing Ctrl+Shift+F

Give feedback

Microsoft Azure

Home > Compute infrastructure | Virtual machines

Compute infrastructure | Virtual machines

Search resources, services, and docs (G+/I)

Help me copy this VM in any region | Manage this VM with Azure CLI

Search

Overview | Activity log | Access control (IAM) | Tags | Diagnose and solve problems | Monitor | Resource visualizer | Connect | Networking | Settings | Availability + scale | Security | Backup + disaster recovery | Operations | Monitoring | Automation | Help

Virtual Machine Scale Set (VMSS) | Compute Fleet | Arc Machines

Disks + images | Capacity + placement | Related services | Monitoring + Operations | Help

+ Create | Reservations

Filter for any field... | Name ? | vm-a-tickets | vm-b-app | vm-c-broadcast

Showing 1 - 3 of 3. Display count: auto

Add or remove favorites by pressing Ctrl+Shift+F

vm-a-tickets

Virtual machine

Help me copy this VM in any region | Manage this VM with Azure CLI

Advisor (1 of 3): Enable Trusted Launch foundational excellence, and modern security for Existing Generation 2 VM(s) ->

Help me copy this VM in any region

Connect | Start | Restart | Stop | Hibernate | Capture | Delete | Refresh | Scale

Properties | Monitoring | Capabilities (7) | Recommendations (3) | Tutorials

Virtual machine

Computer name	vm-a-tickets
Operating system	Linux (ubuntu 22.04)
VM generation	V2
VM architecture	x64
Agent status	Ready
Agent version	2.15.1.3
Hibernation	Disabled
Host group	-
Host	-
Proximity placement group	-
Colocation status	N/A
Capacity reservation group	-
Disk controller type	SCSI

Networking

Public IP address	20.198.112.99 (Network interface vm-a-tickets/VMNic)
Public IP address (IPv6)	-
Private IP address	10.0.1.4
Private IP address (IPv6)	-
Virtual network/subnet	CityNet/subnet-a
DNS name	Configure

Size

Size	Standard B2ats v2
vCPUs	2
RAM	1 GiB
Threads per core	2

Azure Spot

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Microsoft Azure

Home > Compute infrastructure > Virtual machines

Compute infrastructure | Virtual machines

Search

Virtual machines

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Monitor

Resource visualizer

Connect

Networking

Settings

Availability + scale

Security

Backup + disaster recovery

Operations

Monitoring

Automation

Help

Showing 1 - 3 of 3. Display count: auto

vm-b-app

Virtual machine

Help me copy this VM in any region

Manage this VM with Azure CLI

Advisor (1 of 3): Migrate workload to D-series or better virtual machine ->

Help me copy this VM in any region

Connect > Start > Restart > Stop > Hibernate > Capture > Delete > Refresh > Scale

Properties

Monitoring

Capabilities (7)

Recommendations (3)

Tutorials

Virtual machine

Computer name: vm-b-app

Operating system: Linux (ubuntu 22.04)

VM generation: V2

VM architecture: x64

Agent status: Ready

Agent version: 2.15.1.3

Hibernation: Disabled

Host group: -

Host: -

Proximity placement group: -

Colocation status: N/A

Capacity reservation group: -

Disk controller type: SCSI

Networking

Public IP address: 20.204.102.238 (Network interface: vm-b-app/VMNIC)

1 associated public IPs

Public IP address (IPv6): -

Private IP address: 10.0.2.4

Private IP address (IPv6): -

Virtual network/subnet: CityNet/subnet-b

DNS name: Configure

Size

Size: Standard B2ats v2

vCPUs: 2

RAM: 1 GiB

Threads per core: 2

Azure Spot

Microsoft Azure

Home > Compute infrastructure > Virtual machines

Compute infrastructure | Virtual machines

Search

Virtual machines

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Diagnose and solve problems

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Resource visualizer

Connect

Networking

Settings

Availability + scale

Security

Backup + disaster recovery

Operations

Monitoring

Automation

Help

Showing 1 - 3 of 3. Display count: auto

vm-c-broadcast

Virtual machine

Help me copy this VM in any region

Manage this VM with Azure CLI

Advisor (1 of 3): Migrate workload to D-series or better virtual machine ->

Help me copy this VM in any region

Connect > Start > Restart > Stop > Hibernate > Capture > Delete > Refresh > Scale

Properties

Monitoring

Capabilities (7)

Recommendations (3)

Tutorials

Virtual machine

Computer name: vm-c-broadcast

Operating system: Linux (ubuntu 22.04)

VM generation: V2

VM architecture: x64

Agent status: Ready

Agent version: 2.15.1.3

Hibernation: Disabled

Host group: -

Host: -

Proximity placement group: -

Colocation status: N/A

Capacity reservation group: -

Disk controller type: SCSI

Networking

Public IP address: 20.219.219.217 (Network interface: vm-c-broadcast/VMNIC)

1 associated public IPs

Public IP address (IPv6): -

Private IP address: 10.0.3.4

Private IP address (IPv6): -

Virtual network/subnet: CityNet/subnet-c

DNS name: Configure

Size

Size: Standard B2ats v2

vCPUs: 2

RAM: 1 GiB

Threads per core: 2

Azure Spot

5. PUBLIC IP ADDRESSES :-

Public IP addresses were assigned to all Virtual Machines for secure remote management and SSH connectivity. These IPs allow administrative access and testing from external networks.

Microsoft Azure

Search resources, services, and docs (G+I)

Home > Network foundation

Network foundation | Public IP addresses

Review resiliency of public IP addresses | Ensure high availability for public IP resources

Search

Overview

Virtual network

- Virtual Network overview
- Virtual networks
- NAT gateways
- Public IP addresses
- Network interfaces
- Network security groups
- Application security groups
- Bastions
- Route tables
- Route servers

Private Link

DNS

Monitoring and management

Filter for any field...

Subscription equals all

Resource Group equals all

Location equals all

+ Add filter

Name	IP address	IP version	SKU	Associated to
vm-a-ticketsPublicIP	20.198.112.99	IPv4	Standard	vm-a-ticketsVMNIC
vm-b-appPublicIP	20.204.102.238	IPv4	Standard	vm-b-appVMNIC
vm-c-broadcastPublicIP	20.219.219.217	IPv4	Standard	vm-c-broadcastVMNIC

Showing 1 - 3 of 3. Display count: auto

Add or remove favorites by pressing Ctrl+Shift+F

Give feedback

6. NETWORK SECURITY GROUP (NSG) :-

NSG → Inbound security rules

Network Security Groups were configured to control inbound and outbound traffic between Virtual Machines. Custom ICMP and security rules were implemented to allow and block communication based on project requirements.

Microsoft Azure

Search resources, services, and docs (G+I)

Home > Network foundation

Network foundation | Network security groups

Check security rule conflicts across network security groups | Suggest a firewall SKU for my network topology

Search

Overview

Virtual network

- Virtual Network overview
- Virtual networks
- NAT gateways
- Public IP addresses
- Network interfaces
- Network security groups
- Application security groups
- Bastions
- Route tables
- Route servers

Private Link

DNS

Monitoring and management

Filter for any field...

Subscription equals all

Resource Group equals all

Location equals all

+ Add filter

Name	Resource Group	Location	Subscription	Flow log
vm-a-ticketsNSG	rg-stadium-sunil	Central India	Azure for Students	
vm-b-appNSG	rg-stadium-sunil	Central India	Azure for Students	
vm-c-broadcastNSG	rg-stadium-sunil	Central India	Azure for Students	

Showing 1 - 3 of 3. Display count: auto

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Microsoft Azure

Search resources, services, and docs (G+/I)

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Home > Network foundation | Network security groups

vm-a-ticketsNSG
Network security group

How do I create an alert to track firewall metric failures

Search

Move Delete Refresh Give feedback

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

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Automation

Help

Essentials

Resource group (move) : rg-stadium-sunil

Location : Central India

Subscription (move) : Azure for Students

Subscription ID : 32ca749f-f8b6-42b6-8c5a-f71280743ce1

Tags (edit) : Add tags

Custom security rules : 1 inbound, 0 outbound

Associated with : 0 subnets, 1 network interfaces

Filter by name

Port == all Protocol == all Source == all Destination == all Action == all

Inbound Security Rules

Priority ↑↓	Name ↑↓	Port ↑↓	Protocol ↑↓	Source ↑↓	Destination ↑↓	Action ↑↓	
1000	default-allow-ssh	22	Tcp	Any	Any	Allow	
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow	
65001	AllowAzureLoadBalancerInBo...	Any	Any	AzureLoadBalancer	Any	Allow	
65500	DenyAllInBound	Any	Any	Any	Any	Deny	

Outbound Security Rules

65000	AllowVnetOutBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow	
65001	AllowInternetOutBound	Any	Any	Any	Internet	Allow	
65500	DenyAllOutBound	Any	Any	Any	Any	Deny	

Ad Add or remove favorites by pressing CTRL+SHIFT+F

Microsoft Azure

Search resources, services, and docs (G+/I)

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vm-b-appNSG
Network security group

How do I create an alert to track firewall metric failures

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Move Delete Refresh Give feedback

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Essentials

Resource group (move) : rg-stadium-sunil

Location : Central India

Subscription (move) : Azure for Students

Subscription ID : 32ca749f-f8b6-42b6-8c5a-f71280743ce1

Tags (edit) : Add tags

Custom security rules : 2 inbound, 0 outbound

Associated with : 0 subnets, 1 network interfaces

Filter by name

Port == all Protocol == all Source == all Destination == all Action == all

Inbound Security Rules

Priority ↑↓	Name ↑↓	Port ↑↓	Protocol ↑↓	Source ↑↓	Destination ↑↓	Action ↑↓	
100	allow-a-to-b	Any	Icmp	10.0.1.0/24	Any	Allow	
1000	default-allow-ssh	22	Tcp	Any	Any	Allow	
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow	
65001	AllowAzureLoadBalancerInBo...	Any	Any	AzureLoadBalancer	Any	Allow	
65500	DenyAllInBound	Any	Any	Any	Any	Deny	

Outbound Security Rules

65000	AllowVnetOutBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow	
65001	AllowInternetOutBound	Any	Any	Any	Internet	Allow	
65500	DenyAllOutBound	Any	Any	Any	Any	Deny	

Ad Add or remove favorites by pressing CTRL+SHIFT+F

Priority	Name	Port	Protocol	Source	Destination	Action
Inbound Security Rules						
110	allow-b-to-c	Any	Icmp	10.0.2.0/24	Any	Allow
120	deny-a-to-c	Any	Icmp	10.0.1.0/24	Any	Deny
1000	default-allow-ssh	22	Tcp	Any	Any	Allow
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65001	AllowAzureLoadBalancerInBound	Any	Any	AzureLoadBalancer	Any	Allow
65500	DenyAllInBound	Any	Any	Any	Any	Deny
Outbound Security Rules						
65000	AllowVnetOutBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65001	AllowInternetOutBound	Any	Any	Any	Internet	Allow
65500	DenyAllOutBound	Any	Any	Any	Any	Deny

7. A → B PING SUCCESS :-

ping 10.0.2.4

The successful ping from VM-A to VM-B demonstrates allowed internal communication between the Ticketing Server and Application Server through Azure networking and NSG rules.

```
Last login: Thu May 14 04:56:51 2026 from 152.58.78.71
azureuser@vm-a-tickets:~$ ping 10.0.2.4
PING 10.0.2.4 (10.0.2.4) 56(84) bytes of data:
64 bytes from 10.0.2.4: icmp_seq=1 ttl=64 time=1.04 ms
64 bytes from 10.0.2.4: icmp_seq=2 ttl=64 time=1.18 ms
64 bytes from 10.0.2.4: icmp_seq=3 ttl=64 time=0.979 ms
64 bytes from 10.0.2.4: icmp_seq=4 ttl=64 time=1.28 ms
64 bytes from 10.0.2.4: icmp_seq=5 ttl=64 time=1.38 ms
64 bytes from 10.0.2.4: icmp_seq=6 ttl=64 time=1.17 ms
64 bytes from 10.0.2.4: icmp_seq=7 ttl=64 time=1.33 ms
64 bytes from 10.0.2.4: icmp_seq=8 ttl=64 time=1.08 ms
^C
--- 10.0.2.4 ping statistics ---
8 packets transmitted, 8 received, 0% packet loss, time 7009ms
rtt min/avg/max/mdev = 0.979/1.180/1.382/0.135 ms
```

9. B → C PING SUCCESS :-

ping 10.0.3.4

The successful communication between VM-B and VM-C verifies secure application-to-broadcast server connectivity inside the Azure Virtual Network.

```
azureuser@vm-b-app:~$ ping 10.0.3.4
PING 10.0.3.4 (10.0.3.4) 56(84) bytes of data:
64 bytes from 10.0.3.4: icmp_seq=1 ttl=64 time=1.99 ms
64 bytes from 10.0.3.4: icmp_seq=2 ttl=64 time=1.70 ms
64 bytes from 10.0.3.4: icmp_seq=3 ttl=64 time=1.99 ms
64 bytes from 10.0.3.4: icmp_seq=4 ttl=64 time=1.52 ms
64 bytes from 10.0.3.4: icmp_seq=5 ttl=64 time=1.99 ms
64 bytes from 10.0.3.4: icmp_seq=6 ttl=64 time=1.76 ms
64 bytes from 10.0.3.4: icmp_seq=7 ttl=64 time=2.17 ms
^C
--- 10.0.3.4 ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6009ms
rtt min/avg/max/mdev = 1.517/1.873/2.173/0.207 ms
```

10. A → C BLOCKED :-

100% packet loss

Communication between VM-A and VM-C was intentionally blocked using NSG rules. This demonstrates network segmentation and secure communication policies in enterprise cloud infrastructure.

```
azureuser@vm-a-tickets:~$ ping 10.0.3.4
PING 10.0.3.4 (10.0.3.4) 56(84) bytes of data.
^C
--- 10.0.3.4 ping statistics ---
55 packets transmitted, 0 received, 100% packet loss, time 55287ms
```

11. STORAGE ACCOUNT :-

Storage Account → stsunilstadium123

An Azure Storage Account was created to store application data, media files, and structured information securely inside the cloud environment.

Microsoft Azure

Storage center | Blob Storage

Default Directory (sunilgenaraguntla@gmail.onmicrosoft.com)

Search resources, services, and docs (G+)

Which storage accounts have been modified recently? Which storage accounts have unused or inactive containers? Generate a guide for using Azure storage accounts.

Summary Resources

Filter for any field...

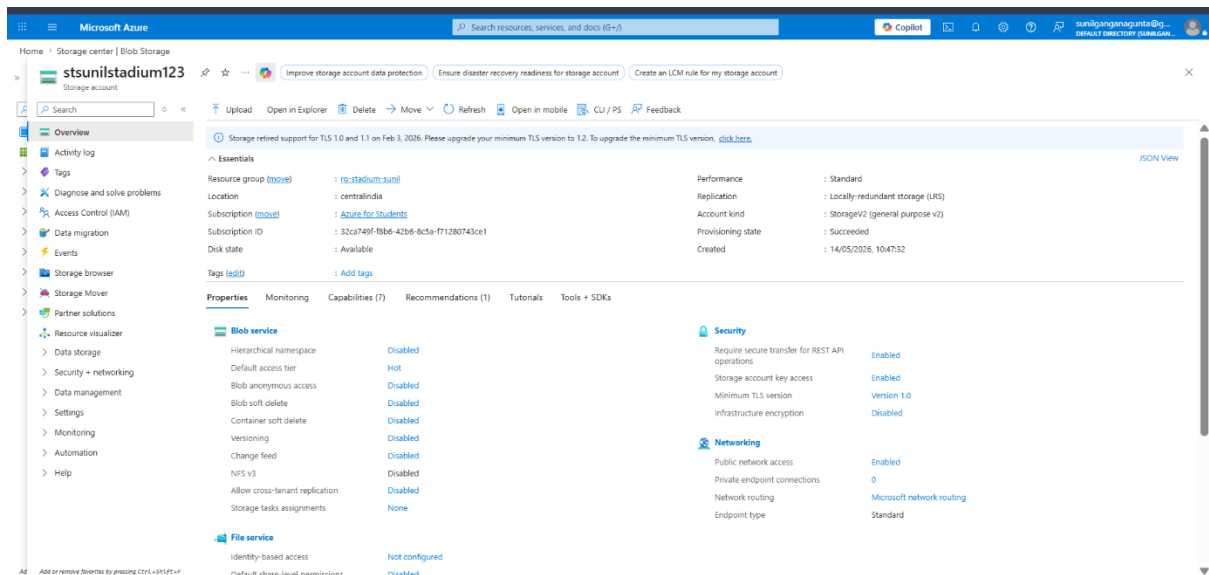
Subscription equals all Resource Group equals all Location equals all Add filter

Name	Type	Kind	Resource Group	Location	Subscription
stsunilstadium123	Storage account	StorageV2	rg-stadium-sunil	Central India	Azure for Students

Showing 1 - 1 of 1. Display count: auto

Add or remove favorites by pressing Ctrl+Shift+F

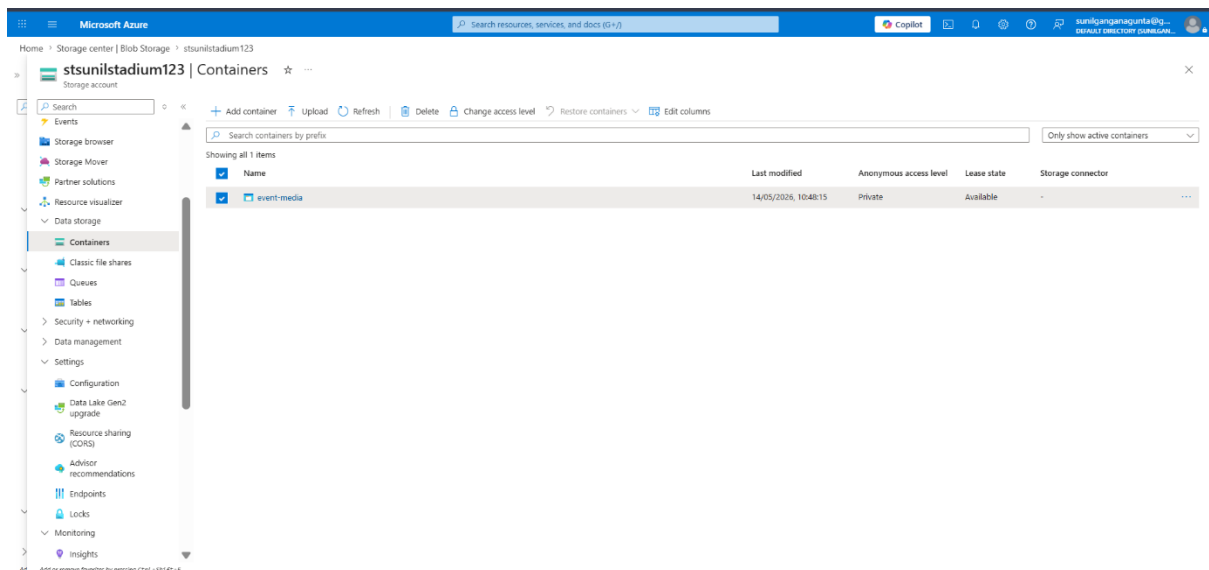
Give feedback



12. BLOB CONTAINER :-

Containers → event-media

The Blob Container named event-media stores media files such as images, videos, and event-related content for the stadium management application.

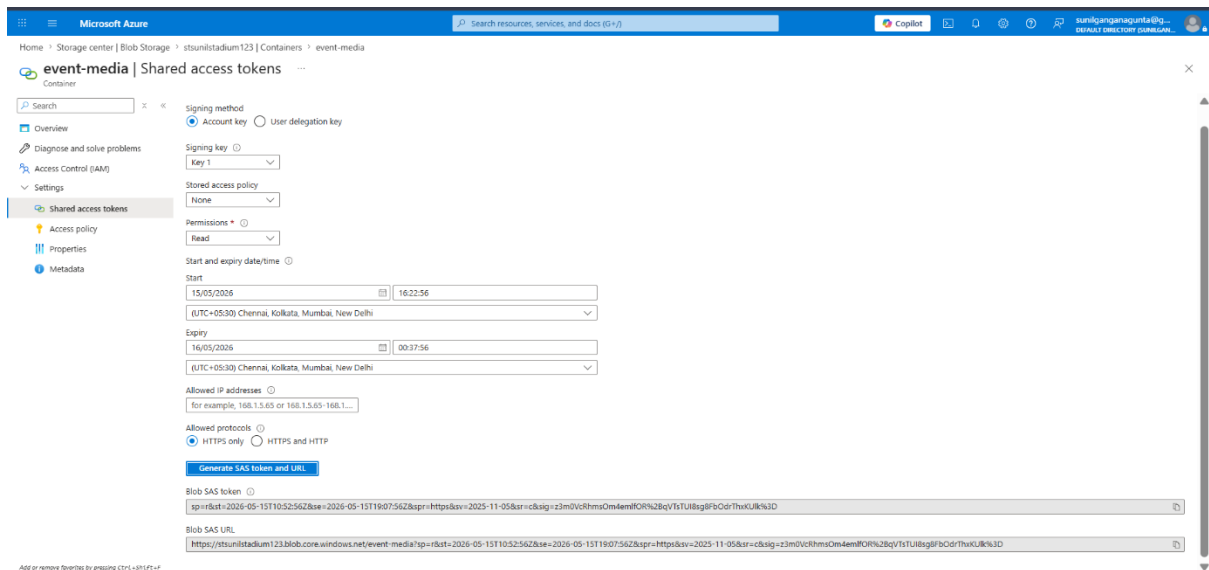


13. SAS TOKEN :-

Generate SAS command + generated token

A Shared Access Signature (SAS) Token was generated to provide temporary and secure access to storage resources without exposing the primary storage account credentials.

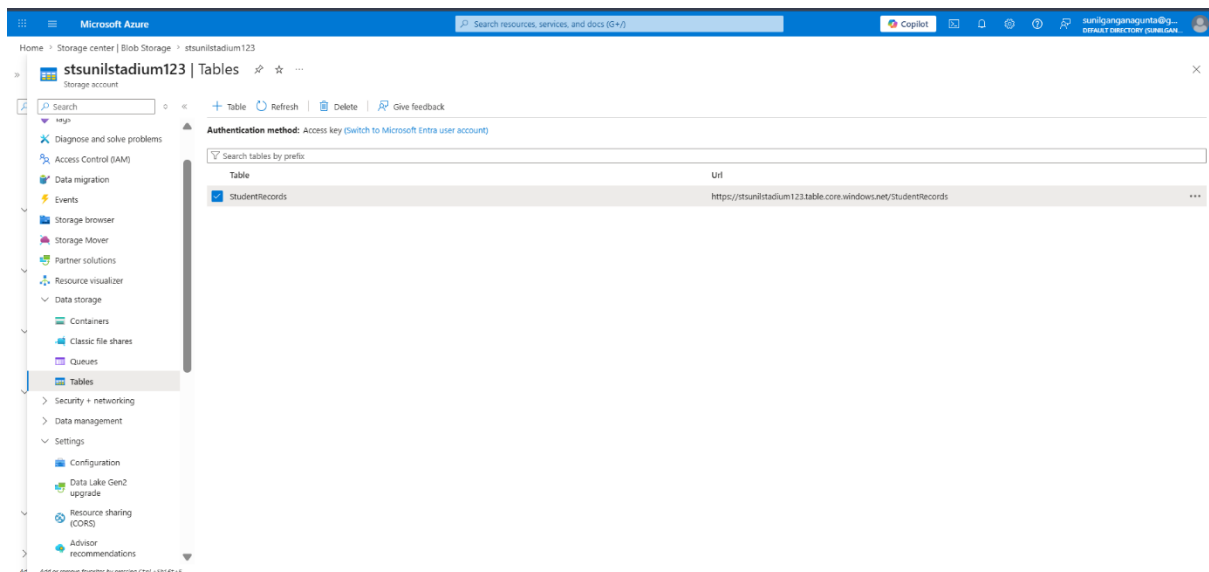
```
C:\Users\sunil>az storage container generate-sas --account-name stsunilstadium123 --name event-media --permissions rl --expiry 2026-05-20T23:59Z --https-only --auth-mode login --as-user "se=2026-05-20T23%3A59Z&sp=rl&spr=https&sv=2026-02-06&sr=c&skoid=0e62ee9f-bdaf-41db-a065-a5113cf19698&sktid=2a063feb-b549-4f0d-a75a-56a12869e9ec&skt=2026-05-14T05%3A23%3A11Z&ske=2026-05-20T23%3A59%3A00Z&skvs=b&skv=2026-02-06&sig=tUUIVZT/ShnGiHCAQj51lxC%2BHPITLJ1H87MmLxrwfZo%3D"
```



14. TABLE STORAGE :-

Tables → StudentRecords

Azure Table Storage was used to store structured data such as records, logs, and application-related information efficiently.



15. USERS :-

Microsoft Entra ID → Users

Internal and external users were created and managed using Microsoft Entra ID to simulate enterprise identity and access management.

Microsoft Azure

Home > Default Directory | Users

Users

Azure Active Directory is now Microsoft Entra ID.

6 users found (6 users selected)

Display name	User principal name	User type	Is Agent	On-premises sy...	Identities	Company name	Creation type
David Chen	david.chen@sunil...	Member	No	No	sunilganganaguntagmail.onmic...		
Internal User	internaluser@sunil...	Member	No	No	sunilganganaguntagmail.onmic...		
James Wilson	James.Wilson@su...	Member	No	No	sunilganganaguntagmail.onmic...		
Lisa Brown	lisa.brown@sunilg...	Member	No	No	sunilganganaguntagmail.onmic...		
Robert Taylor	robert.taylor@sun...	Member	No	No	sunilganganaguntagmail.onmic...		
Sunil Ganganagunta	sunilganganagunt...	Member	No	No	MicrosoftAccount		Invitation

Add or remove favorites by pressing Ctrl+Shift+F

16. GROUPS :-

Microsoft Entra ID → Groups

Azure Groups were used to organize users and simplify permission management for cloud resources and administrative operations.

Microsoft Azure

Home > Default Directory | Groups

Groups | Overview

Overview

Search your tenant

Basic information

Total groups	1	Dynamic groups	0
M365 groups	0	Cloud groups	1
Security groups	1	On-premises groups	0

Alerts

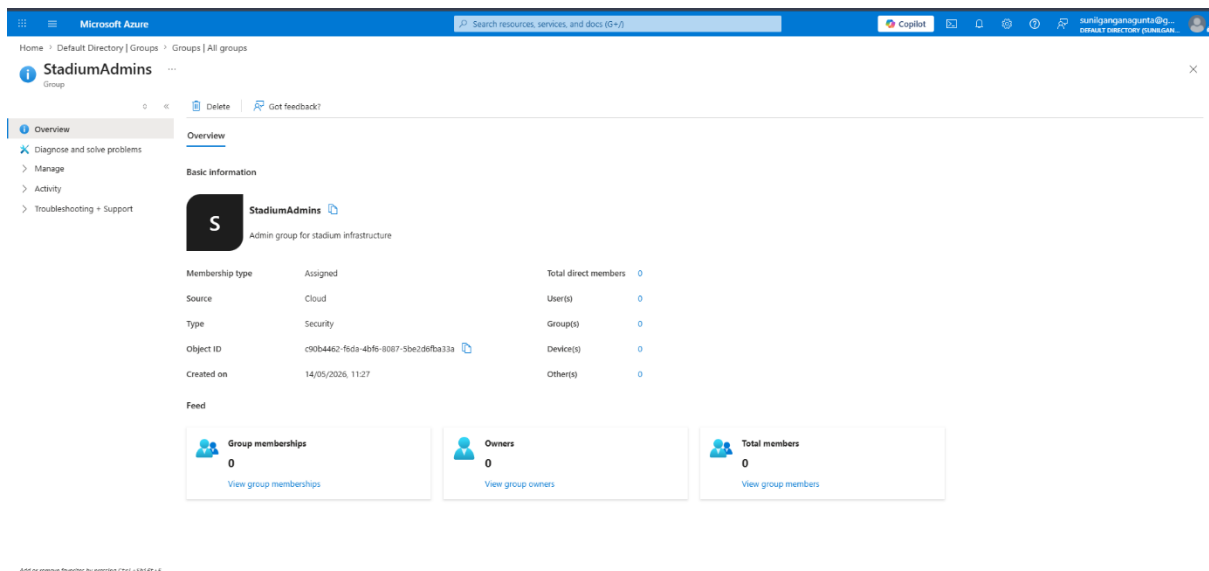
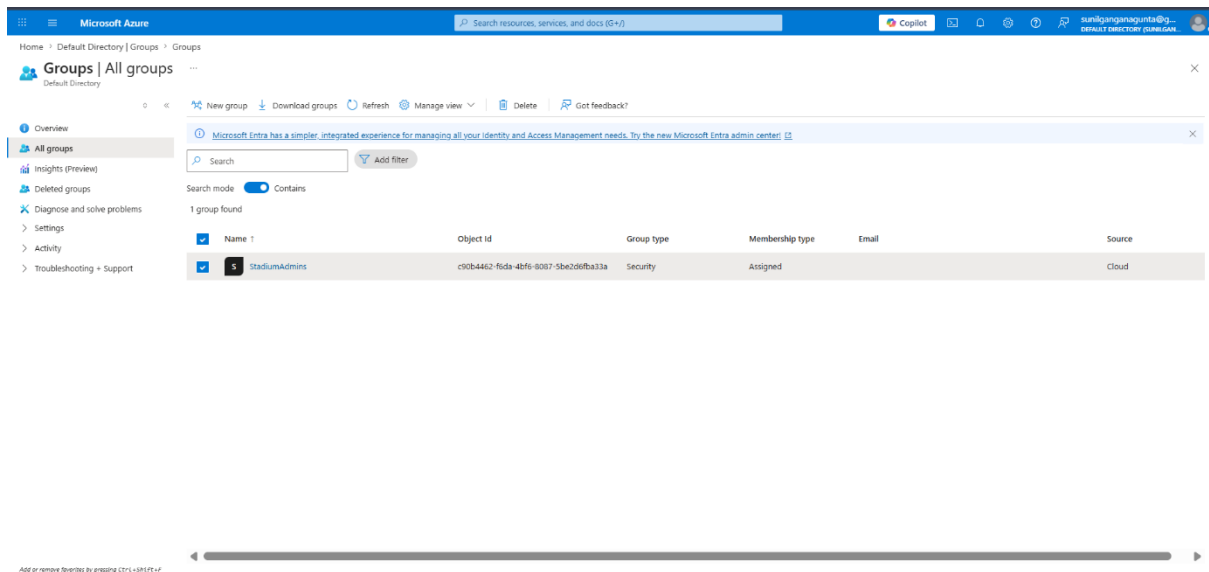
Feature highlights

Access reviews
Make sure only the right people have continued access.

Quick actions

Add group Download groups

Add or remove favorites by pressing Ctrl+Shift+F



17. RBAC ROLE ASSIGNMENT :-

Access Control (IAM)

Role-Based Access Control (RBAC) was implemented to assign specific permissions such as Reader and Contributor roles to users and groups.

Microsoft Azure

Home > Resource Manager > Resource groups > rg-stadium-sunil

rg-stadium-sunil | Access control (IAM)

Search resources, services, and docs (G+V)

Current role assignments

Eligible assignments

Assignments for the selected user, group, service principal, or managed identity at this scope or inherited to this scope.

Search by assignment name or description

Role	Description	Scope	Group assignment	Condition
Owner	Grants full access to manage all r...	Subscription (Inherited)	--	None
Owner	Grants full access to manage all r...	Subscription (Inherited)	--	None
Reader	View all resources, but does not ...	This resource	--	None

Deny assignments (0)

Check access

Role assignments

Roles

Deny assignments

My access

View my level of access to this resource.

View my access

Check access

Review the level of access a user, group, service principal, or managed identity has to this resource. [Learn more](#)

Grant access to this resource

Grant access to resources by assigning a role. [Learn more](#)

Add role assignment

View access to this resource

View the role assignments that grant access to this and other resources. [Learn more](#)

View

View deny assignments

View the role assignments that deny access to this and other resources. [Learn more](#)

View

18. AZURE POLICY :-

Policy → Assignments

Microsoft Azure

Home > Policy

Policy | Assignments

Assign policy Assign initiative Refresh

Search

Filter by name or ID...

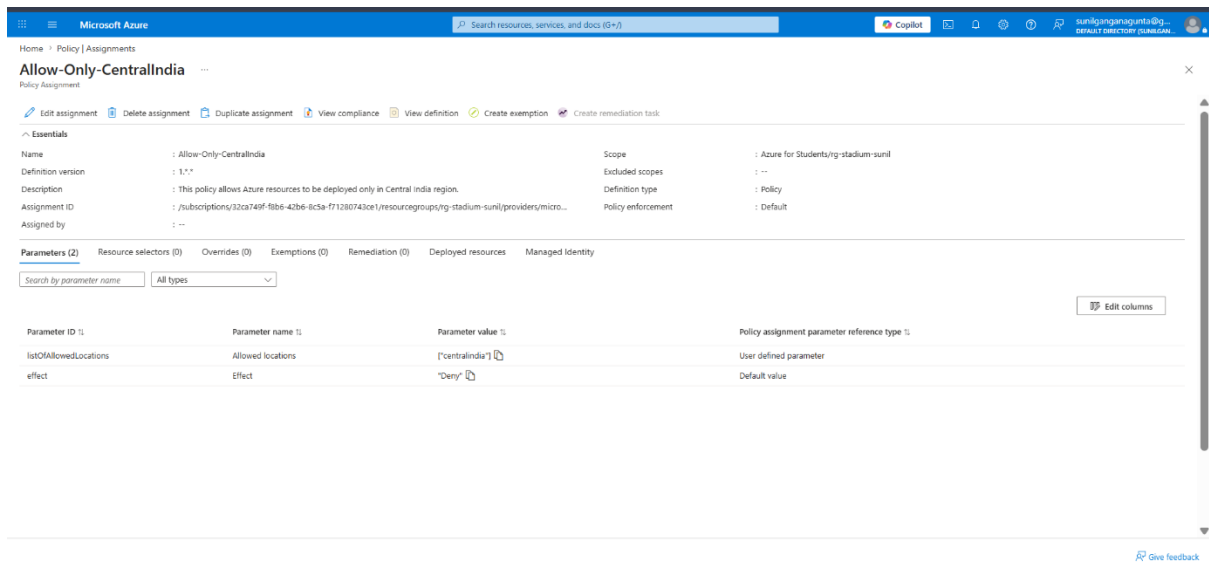
Scope: Azure for Students Definition type: All definition types

Total Assignments 3 Initiative Assignments 1 Policy Assignments 2

Assignment name	Scope	Type
Allowed resource deployment regions	Azure for Students	Policy
ASC Default (subscriptions: 32ca749f-88b6-42b6-8c5e-f71280743ce1)	Azure for Students	Initiative
Allow-Only-CentralIndia	Azure for Students/rg-stadium-sunil	Policy

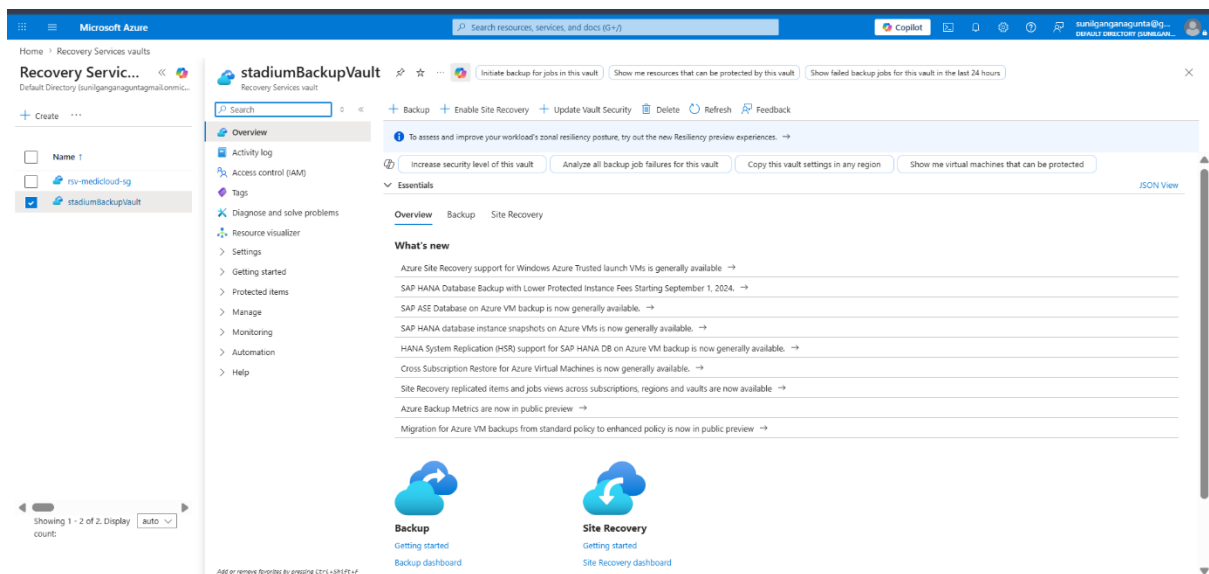
Edit columns Export to CSV

Give feedback



19. BACKUP VAULT :

Recovery Services Vault → stadiumBackupVault



20. LOAD BALANCER :

Load Balancer → stadiumLB

Azure Load Balancer distributes incoming traffic across backend resources to improve availability, scalability, and reliability of the cloud application.

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

sunilgangaragunta@...
DEFAULT DIRECTORY (SUNILGAN...

Home > Load balancing and content delivery > Load balancers

Load balancing and content delivery ...
Preview

Search

+ Create Manage view

Overview

Load balancer

Application gateways

Load balancers

Content delivery

DNS load balancing

Related services

Name 1

stadiumLB

stadiumLB

Load balancer

Run diagnostic for my Load Balancer Troubleshoot connectivity issues in this load balancer Analyze traffic flow through this load balancer

Search

Move Delete Refresh Give feedback

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Monitoring

Automation

Help

Essentials

Resource group (...): rg-stadium-sunil

Location (move): Central India

Subscription (move): Azure for Students

Subscription ID: 32ca749f-18b6-42b6-8c5a-f71280743ce1

SKU: Standard

Tier: Regional

Tags (add): Add tags

Backend pool: stadiumBackend (2 virtual machines)

Load balancing rule: http-rule (Tcp/80 to Tcp/79)

Health probe: healthProbe (Tcp/80)

Inbound NAT rules: -

Frontend IP address: 10.0.2.5

Configure high availability and scalability for your applications

Create highly-available and scalable applications in minutes by using built-in load balancing for cloud services and virtual machines. Azure Load Balancer supports TCP/UDP-based protocols and protocols used for real-time voice and video messaging applications. [Learn more](#)

Balance IPv4 and IPv6 addresses

Native dual-stack endpoints help meet regulatory requirements and address the fast-growing number of devices in mobile and IoT. [Learn more](#)

View frontend IP configuration

View backend pools

Build highly reliable applications

Load Balancer improves application uptime by routing traffic to healthy nodes. [Learn more](#)

View health probes

View load balancing rules

Secure your networks

Control network traffic and protect private networks using built-in network address translation (NAT). [Learn more](#)

View inbound NAT rules

Showing 1 - 1 of 1. Display count: auto

Add or remove favorites by pressing CTRL+SHIFT+F

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

sunilgangaragunta@...
DEFAULT DIRECTORY (SUNILGAN...

Home > Network foundation > Virtual networks > CityNet

Network foundation | Virtual networks ...
Preview

Search

+ Create Manage view

Overview

Virtual network

Virtual Network overview

Virtual networks

NAT gateways

Public IP addresses

Network interfaces

Network security groups

Application security groups

Bastions

Route tables

Route servers

Private Link

DNS

Monitoring and management

Name 1

CityNet

CityNet | Subnets

Virtual network

Subnet Refresh Manage users Delete Export to CSV

Search subnets

Create subnets to segment the virtual network address space into smaller ranges for use by your applications. When you deploy resources into a subnet, Azure assigns the resource an IP address from the subnet.

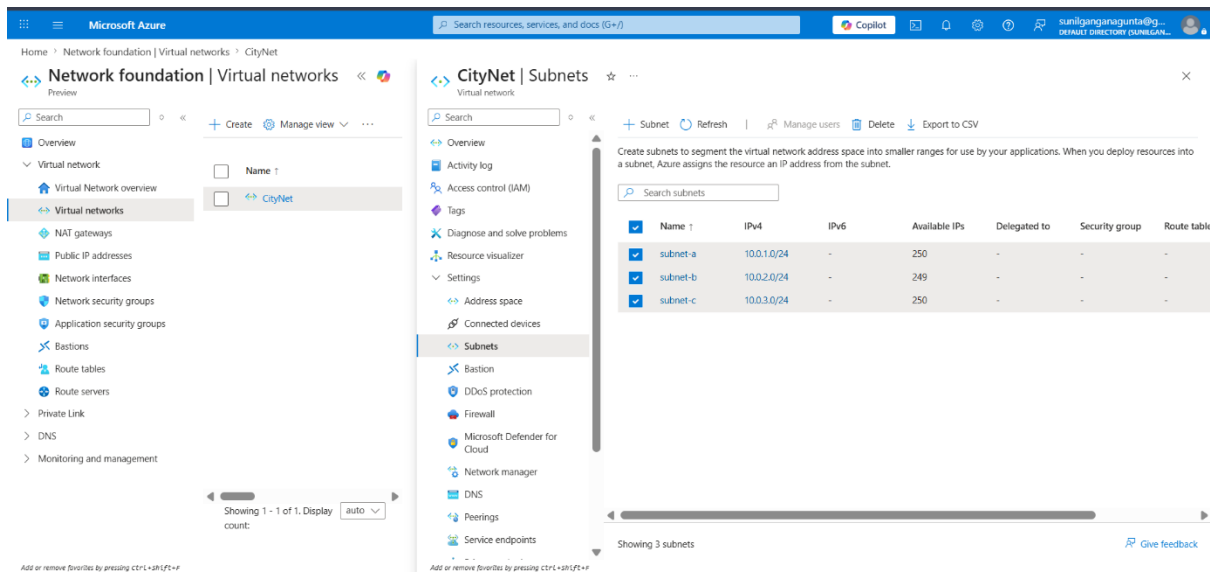
☒	Name ↑	IPv4	IPv6	Available IPs	Delegated to	Security group	Route table
☒	subnet-a	10.0.1.0/24	-	250	-	-	-
☒	subnet-b	10.0.2.0/24	-	249	-	-	-
☒	subnet-c	10.0.3.0/24	-	250	-	-	-

Showing 3 subnets

Give feedback

Showing 1 - 1 of 1. Display count: auto

Add or remove favorites by pressing CTRL+SHIFT+F



21. CONCLUSION :

This project successfully demonstrates the deployment of a secure and scalable multi-tier cloud infrastructure using Microsoft Azure services. The implementation includes networking, virtualization, storage, identity management, governance, and security components commonly used in enterprise cloud environments.